End of Term 1 Exam

Question Booklet

Grade 9 Advanced – Mathematics

Reveal · Academic Year 2026–2027

Name _____ Class _____

EXAM FOCUS

LESSON 2-1 **Writing and Interpreting Equations**

QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q1	MCQ Multiple choice	Translate sentences into equations	**Page 71** · Exercises 1–4

Multiple choice · Choose the equation that represents each sentence.

Textbook · Ex. 1 ●●○

1 Two added to three times a number m is the same as 18.

(A) $3(m+2)=18$ (B) $3m+2=18$

(C) $3+m+2=18$ (D) $3m=2+18$

Textbook · Ex. 2 ●●○

2 The product of five and the sum of a number x and three is twelve.

(A) $5x+3=12$ (B) $5+(x+3)=12$

(C) $5(x+3)=12$ (D) $3(x+5)=12$

Textbook · Ex. 3 ●●○

3 The quotient of 24 and x equals 14 minus 2 times x.

(A) $\frac{x}{24}=14-2x$ (B) $\frac{24}{x}=2x-14$ (C) $24x=14-2x$ (D) $\frac{24}{x}=14-2x$

Textbook · Ex. 4 ●●●

4 Nine times a number y subtracted from 85 is seven times the sum of four and y.

(A) $85-9y=7(4+y)$ (B) $9y-85=7(4+y)$ (C) $85-9y=7\cdot 4+y$ (D) $9(y-85)=7(4+y)$

●○○

5 Eight more than a number n is 21.

(A) $8n=21$ (B) $n-8=21$

(C) $n+8=21$ (D) $n=8+21$

●○○

6 Twelve less than a number k equals 30.

(A) $12-k=30$ (B) $k-12=30$

(C) $k+12=30$ (D) $30-k=12$

●●○

7 Four times a number a decreased by 9 is the same as 23.

(A) $4(a-9)=23$ (B) $9-4a=23$

(C) $4a+9=23$ (D) $4a-9=23$

●●○

8 The quotient of a number p and 6 is equal to five more than p.

(A) $\frac{p}{6}=p+5$ (B) $\frac{6}{p}=p+5$

(C) $\frac{p}{6}=5p$ (D) $\frac{p}{6}+5=p$

●●○

9 Three times the sum of a number w and 8 is 45.

(A) $3w+8=45$ (B) $3+(w+8)=45$

(C) $3(w+8)=45$ (D) $8(w+3)=45$

●●●

10 Five times a number x subtracted from 60 is the same as two times x.

(A) $5x-60=2x$ (B) $60-5x=2x$

(C) $5(60-x)=2x$ (D) $60-5x=x+2$

●●●

11 The difference of 40 and twice a number d equals the quotient of d and 4.

(A) $2d-40=\frac{d}{4}$ (B) $40-2d=\frac{4}{d}$ (C) $2(40-d)=\frac{d}{4}$ (D) $40-2d=\frac{d}{4}$

●●●

12 Six less than the product of 3 and a number z is equal to four times the difference of z and 5.

(A) $3z-6=4(z-5)$ (B) $6-3z=4(z-5)$ (C) $3(z-6)=4(z-5)$ (D) $3z-6=4z-5$

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LESSON 2-3 **Solving Multi-Step Equations**

QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q2	MCQ Multiple choice	Solve equations involving more than one operation	**Page 88** · Exercises 1–12

Multiple choice · Solve each equation, then choose the correct solution.

Textbook · Ex. 1 ●○○

1 $3t + 7 = -8$

(A) $t = 5$ (B) $t = -\frac{1}{3}$ (C) $t = -5$ (D) $t = -45$

Textbook · Ex. 2 ●○○

2 $8 = 16 + 8n$

(A) $n = -1$ (B) $n = 1$ (C) $n = 3$ (D) $n = -8$

Textbook · Ex. 3 ●○○

3 $-34 = 6m - 4$

(A) $m = -\frac{19}{3}$ (B) $m = -5$ (C) $m = 5$ (D) $m = -30$

Textbook · Ex. 4 ●○○

4 $9x + 27 = -72$

(A) $x = -99$ (B) $x = 11$ (C) $x = -5$ (D) $x = -11$

Textbook · Ex. 5 ●●○

5 $\frac{y}{5} - 6 = 8$

(A) $y = 10$ (B) $y = 70$ (C) $y = \frac{14}{5}$ (D) $y = -70$

Textbook · Ex. 6 ●●○

6 $\frac{f}{-7} - 8 = 2$

(A) $f = -70$ (B) $f = 70$ (C) $f = 42$ (D) $f = -\frac{10}{7}$

Textbook · Ex. 7 ●●○

7 $1 + \frac{r}{9} = 4$

(A) $r = \frac{1}{3}$ (B) $r = -27$ (C) $r = 27$ (D) $r = 45$

Textbook · Ex. 8 ●●○

8 $\frac{k}{3} + 4 = -16$

(A) $k = -36$ (B) $k = -\frac{20}{3}$ (C) $k = 60$ (D) $k = -60$

Textbook · Ex. 9 ●●○

9 $\frac{n - 2}{7} = 2$

(A) $n = 12$ (B) $n = 9$ (C) $n = 16$ (D) $n = -12$

Textbook · Ex. 10 ●●●

10 $14 = \frac{6 + z}{-2}$

(A) $z = 22$ (B) $z = -34$ (C) $z = -22$ (D) $z = 34$

Textbook · Ex. 11 ●●●

11 $-11 = \frac{a - 5}{6}$

(A) $a = -61$ (B) $a = -71$ (C) $a = -1$ (D) $a = 61$

Textbook · Ex. 12 ●●●

12 $\frac{22 - w}{3} = -7$

(A) $w = 1$ (B) $w = -43$ (C) $w = -1$ (D) $w = 43$

●●○

13 $7 - 2x = 15$

(A) $x = 4$ (B) $x = -11$ (C) $x = -4$ (D) $x = 11$

●●●

14 $\frac{x + 4}{3} = -5$

(A) $x = -19$ (B) $x = -11$ (C) $x = -9$ (D) $x = 11$

●●○

15 $-2 = \frac{m}{4} + 6$

(A) $m = 16$ (B) $m = -2$ (C) $m = 32$ (D) $m = -32$

●●●

16 $5 - \frac{k}{2} = 9$

(A) $k = 28$ (B) $k = -8$ (C) $k = 8$ (D) $k = -2$

●●○

17 $3 - 4y = 23$

(A) $y = 5$ (B) $y = -\frac{13}{2}$ (C) $y = -5$ (D) $y = -80$

●●●

18 $\frac{2x - 1}{5} = 3$

(A) $x = 7$ (B) $x = 8$ (C) $x = 2$ (D) $x = -7$

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LESSON 2-4 Solving Equations with the Variable on Each Side

QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q3	MCQ Multiple choice	Solve equations with the variable on each side	**Page 97** · Exercises 1–6

Multiple choice · Solve each equation, then choose the correct solution.

Textbook · Ex. 1 ●○○

1 $7c + 12 = -4c + 78$

(A) $c = 22$ (B) $c = -6$ (C) $c = 6$ (D) $c = \frac{90}{11}$

Textbook · Ex. 2 ●○○

2 $2m - 13 = -8m + 27$

(A) $m = 4$ (B) $m = -\frac{20}{3}$ (C) $m = -4$ (D) $m = \frac{7}{5}$

Textbook · Ex. 3 ●○○

3 $9x - 4 = 2x + 3$

(A) $x = \frac{7}{11}$ (B) $x = -1$ (C) $x = 7$ (D) $x = 1$

Textbook · Ex. 4 ●●○

4 $6 + 3t = 8t - 14$

(A) $t = -4$ (B) $t = 4$ (C) $t = \frac{8}{5}$ (D) $t = \frac{20}{11}$

Textbook · Ex. 5 ●●●

5 $\frac{b - 4}{6} = \frac{b}{2}$

(A) $b = 2$ (B) $b = -\frac{4}{5}$ (C) $b = -2$ (D) $b = 4$

Textbook · Ex. 6 ●●●

6 $\frac{3v + 12}{6} = \frac{4v}{3}$

(A) $v = \frac{12}{5}$ (B) $v = 12$ (C) $v = 4$ (D) $v = -\frac{12}{5}$

7 ●○○ $5x + 3 = 2x + 18$

(A) $x = 3$ (B) $x = 5$ (C) $x = 7$ (D) $x = -5$

8 ●○○ $4y - 7 = y + 11$

(A) $y = \frac{18}{5}$ (B) $y = -6$ (C) $y = \frac{4}{3}$ (D) $y = 6$

9 ●●○ $3a + 10 = 8a - 5$

(A) $a = 3$ (B) $a = 1$ (C) $a = -3$ (D) $a = \frac{5}{11}$

10 ●●○ $12 - n = 3n + 4$

(A) $n = 4$ (B) $n = -2$ (C) $n = 2$ (D) $n = 8$

11 ●●○ $9k + 5 = 4k - 20$

(A) $k = 5$ (B) $k = -5$ (C) $k = -3$ (D) $k = -\frac{25}{13}$

12 ●●○ $-2p + 9 = 3p - 11$

(A) $p = 2$ (B) $p = -4$ (C) $p = 20$ (D) $p = 4$

13 ●●○ $8 - 3x = 2x - 17$

(A) $x = 5$ (B) $x = -9$ (C) $x = -5$ (D) $x = 25$

14 ●●○ $11d - 4 = 5d + 26$

(A) $d = -5$ (B) $d = \frac{11}{8}$ (C) $d = 5$ (D) $d = 4$

15 ●●● $\frac{x}{3} = \frac{x + 8}{5}$

(A) $x = 4$ (B) $x = 3$ (C) $x = 12$ (D) $x = -12$

16 ●●● $\frac{y + 2}{4} = \frac{y}{2}$

(A) $y = -2$ (B) $y = 2$ (C) $y = 4$ (D) $y = 1$

17 ●●● $\frac{2m - 6}{4} = \frac{m}{3}$

(A) $m = 3$ (B) $m = -9$ (C) $m = 6$ (D) $m = 9$

18 ●●● $6 - 4c = 2c + 6$

(A) $c = 0$ (B) $c = 2$ (C) $c = 1$ (D) $c = -2$

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LESSON 2-4 **Solving Equations with the Variable on Each Side**

QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q21	FRQ Free response	Solve equations by applying the Distributive Property	**Page 97** · Exercises 7–16

Written answer · Solve each equation. Show every step, then write your final answer in the box.

Textbook · Ex. 7 ●○○

1 Solve $2(r+6)=4(r+4)$.

SOLUTION

Final answer

Textbook · Ex. 8 ●○○

2 Solve $6(n+5)=3(n+16)$.

SOLUTION

Final answer

Textbook · Ex. 9 ●●○

3 Solve $5(g+8)-7=117-g$.

SOLUTION

Final answer

Textbook · Ex. 10 ●●●

4 Solve $12-\frac{4}{5}(x+15)=\frac{2}{5}x+6$.

SOLUTION

Final answer

Textbook · Ex. 11 ●●○

5 Solve $3(3m-2)=2(3m+3)$.

SOLUTION

Final answer

Textbook · Ex. 12 ●●○

6 Solve $6(3a+1)-30=3(2a-4)$.

SOLUTION

Final answer

Textbook · Ex. 13 ●○○

7 Solve $7n + 6 = 4n - 9$.

SOLUTION

Final answer

Textbook · Ex. 14 ●●●

8 Solve $-6(2r + 8) = -10(r - 3)$.

SOLUTION

Final answer

Textbook · Ex. 15 ●●●

9 Solve $5 - 3(w + 4) = w - 7$.

SOLUTION

Final answer

Textbook · Ex. 16 ●●●

10 Solve $2x - 5(x - 3) = 2(x - 10)$.

SOLUTION

Final answer

●●○

11 Solve $4(x - 2) = 2(x + 5)$.

SOLUTION

Final answer

●●●

12 Solve $3 - 2(y + 1) = 4y - 11$.

SOLUTION

Final answer

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LESSON 2-5 Solving Equations Involving Absolute Value

QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q4	MCQ Multiple choice	Solve absolute value equations	**Page 106** · Exercises 19–23

Multiple choice · Write the absolute value equation and solve it, then choose the correct answer.

Textbook · Ex. 19 ●○○

1 An oven is set to 400°F with a tolerance of ±15°F. Write and solve an absolute value equation. What are the minimum and maximum temperatures?

(A) 385° and 415°F (B) 400° and 415°F (C) 370° and 430°F (D) 385° and 400°F

Textbook · Ex. 20 ●○○

2 A poll is accurate within ±5%, and 38% of voters chose Candidate A. Write and solve an absolute value equation. What are the minimum and maximum percents for Candidate A?

(A) 38% and 43% (B) 33% and 38% (C) 33% and 43% (D) 28% and 48%

Textbook · Ex. 21a ●●○

3 The ACT math mean is 20.9 with a standard deviation of 5.3. Which equation gives the scores one standard deviation from the mean?

(A) $|x - 5.3| = 20.9$ (B) $|x - 20.9| = 5.3$ (C) $|x + 20.9| = 5.3$ (D) $|x - 20.9| = 10.6$

Textbook · Ex. 21b ●●○

4 Using Exercise 21a, what is the range of scores within **two** standard deviations of the mean?

(A) 15.6 to 26.2 (B) 5.3 to 20.9 (C) 18.2 to 23.6 (D) 10.3 to 31.5

Textbook · Ex. 22 ●●●

5 21% of about 40 million students chose “finding new resources”, with a margin of error of ±3%. What is the range of the number of students?

(A) 7.2 to 9.6 million (B) 8.4 to 9.6 million (C) 18 to 24 million (D) 8.1 to 8.7 million

Textbook · Ex. 23 ●●●

6 Bags of rock salt are labeled 35 lb, and their weight is within 8 ounces of the label (16 oz = 1 lb). What are the minimum and maximum weights?

(A) 27 and 43 lb (B) 34.2 and 35.8 lb (C) 34.5 and 35.5 lb (D) 34.92 and 35.08 lb

●○○

7 $|x + 3| = 7$

(A) $\{4\}$ (B) $\{-10, 4\}$ (C) $\{-4, 10\}$ (D) $\{-7, 7\}$

●○○

8 $|2t - 4| = 8$

(A) $\{-6, 2\}$ (B) $\{6\}$ (C) $\{-2, 6\}$ (D) $\{-4, 4\}$

●○○

9 $|y - 5| = -2$

(A) $\{3, 7\}$ (B) $\{7\}$ (C) $\{3\}$ (D) $\varnothing$

●●○

10 $|x - 1| + 2 = 5$

(A) $\{-4, 2\}$ (B) $\{-2, 4\}$ (C) $\{-6, 8\}$ (D) $\{4\}$

●●●

11 $2|n - 4| = 10$

(A) $\{-6, 14\}$ (B) $\{9\}$ (C) $\{-9, 1\}$ (D) $\{-1, 9\}$

●●●

12 $|3x + 1| - 4 = 6$

(A) $\{-\frac{11}{3}, 3\}$ (B) $\{-\frac{7}{3}, \frac{5}{3}\}$ (C) $\{-3, \frac{11}{3}\}$ (D) $\{3\}$

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LESSON 2-6 **Solving Proportions**

QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q5	MCQ Multiple choice	Solve proportions	**Page 113** · Exercises 19–26

Multiple choice · Solve each proportion. If necessary, round to the nearest hundredth.

Textbook · Ex. 19 ●●○

1 $\frac{4v+7}{15} = \frac{6v+2}{10}$

A $v = 0.8$ B $v = 1.25$
C $v = -0.8$ D $v = 2$

Textbook · Ex. 20 ●●○

2 $\frac{9b-3}{9} = \frac{5b+5}{3}$

A $b = 3$ B $b = -3$
C $b = -1$ D $b = -0.5$

Textbook · Ex. 21 ●●○

3 $\frac{2n-4}{5} = \frac{3n+3}{10}$

A $n = 5$ B $n = -11$
C $n = 11$ D $n = 1$

Textbook · Ex. 22 ●●●

4 $\frac{2}{g+6} = \frac{4}{5g+10}$

A $g = -0.67$ B $g = 2$
C $g = 0.5$ D $g = 0.67$

Textbook · Ex. 23 ●○○

5 $\frac{x}{3} = \frac{3x+2}{6}$

A $x = -2$ B $x = 2$
C $x = -0.67$ D $x = 6$

Textbook · Ex. 24 ●●○

6 $\frac{w+3}{7} = \frac{w-1}{8}$

A $w = 31$ B $w = -31$
C $w = 17$ D $w = -3$

Textbook · Ex. 25 ●●●

7 $\frac{4q-3}{5} = \frac{2q+1}{7}$

A $q = 0.33$ B $q = -1.44$
C $q = 1.44$ D $q = 2.6$

Textbook · Ex. 26 ●●●

8 $\frac{5}{7k+4} = \frac{2}{2k-3}$

A $k = 5.75$ B $k = -0.25$
C $k = -1.75$ D $k = -5.75$

●○○

9 $\frac{x}{4} = \frac{x+6}{10}$

A $x = 4$ B $x = -4$
C $x = 6$ D $x = 2.4$

●○○

10 $\frac{a+2}{3} = \frac{a-1}{2}$

A $a = -7$ B $a = 7$
C $a = 1$ D $a = 5$

●●○

11 $\frac{3m-1}{4} = \frac{m+5}{2}$

A $m = 9$ B $m = -11$
C $m = 11$ D $m = 3$

●●○

12 $\frac{6}{y+1} = \frac{3}{y-2}$

A $y = -5$ B $y = 3$
C $y = 1$ D $y = 5$

●●○

13 $\frac{2p+3}{5} = \frac{p-1}{3}$

A $p = -14$ B $p = 14$
C $p = -4$ D $p = 2$

●●○

14 $\frac{5x-2}{6} = \frac{2x+1}{3}$

A $x = 0.67$ B $x = 4$
C $x = -4$ D $x = 2.67$

●●●

15 $\frac{4}{z-3} = \frac{2}{z+1}$

A $z = 5$ B $z = -1$
C $z = -5$ D $z = 3$

●●●

16 $\frac{k+4}{8} = \frac{k-2}{5}$

A $k = -12$ B $k = 1.33$
C $k = 2$ D $k = 12$

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LESSON 2-7 Using Formulas

QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q22	FRQ Free response	Solve equations for specific variables	**Page 120, 126** · Example 4 · Exercises 23–24

Written answer · Answer every part. Show your steps and write the final answers in the box.

Textbook · p. 120 Ex. 4 ●●○

1 **Pizza.** $T = 13.49c + 15.49p$, where c = cheese pizzas and p = pepperoni pizzas.
a. Kishi has \$85. Describe the constraints. **b.** Solve for c. **c.** She buys 3 pepperoni pizzas. Find the maximum number of cheese pizzas.

SOLUTION

Final answer

Textbook · p. 126 Ex. 23 ●●○

2 **Coffee.** $T = 5.50p + 7f$, where p = packages of plain and f = packages of flavored coffee.
a. Consuelo has \$40. Describe the constraints. **b.** Solve for f. **c.** She buys 3 plain packages. Find the maximum number of flavored packages.

SOLUTION

Final answer

Textbook · p. 126 Ex. 24 ●●○

3 **Towels.** $T = 6b + 2w$, where b = bath towels and w = washcloths.
a. The budget is \$85. Describe the constraints. **b.** Solve for b. **c.** She orders 20 washcloths. How many bath towels can she order and stay under budget?

SOLUTION

Final answer

●●○

4 **Fruit.** $T = 3m + 2b$, where m = mangoes (\$3 each) and b = bananas (\$2 each).
a. The budget is \$24. Describe the constraints. **b.** Solve for b. **c.** She buys 5 mangoes. Find the maximum number of bananas.

SOLUTION

Final answer

●○○

5 **a.** Solve $3x + 2y = 12$ for y.
b. Use your answer to find y when $x = 2$.

SOLUTION

Final answer

●●●

6 **Perimeter.** $P = 2\ell + 2w$.
a. Solve for w. **b.** Find the width of a rectangle with $P = 30$ cm and $\ell = 11$ cm.

SOLUTION

Final answer

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LESSON 4-1 Graphing Linear Functions

QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q23	FRQ Free response	Graph linear functions by using the x- and y-intercepts	**Page 216** · Exercises 9–14

Written answer · Find both intercepts, then graph the equation on the grid.

Textbook · Ex. 9 ●○○

1 Graph $y = 4 + 2x$ by using the intercepts.

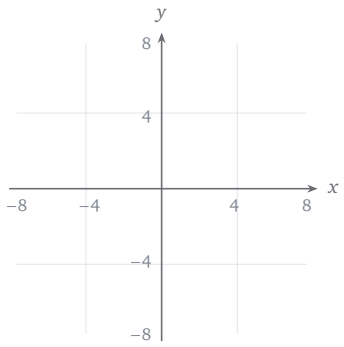
X-INTERCEPT

(_____ , 0)

Y-INTERCEPT

(0 , _____)

WORKING



Textbook · Ex. 10 ●●○

2 Graph $5 - y = -3x$ by using the intercepts.

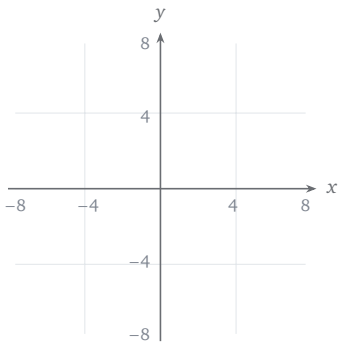
X-INTERCEPT

(_____ , 0)

Y-INTERCEPT

(0 , _____)

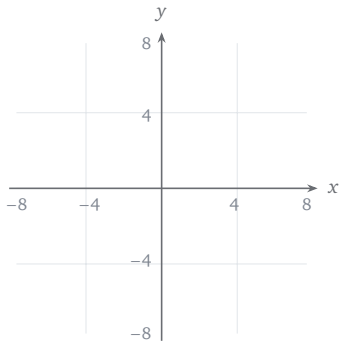
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Textbook · Ex. 11 ●●○

3 Graph $x = 5y + 5$ by using the intercepts.

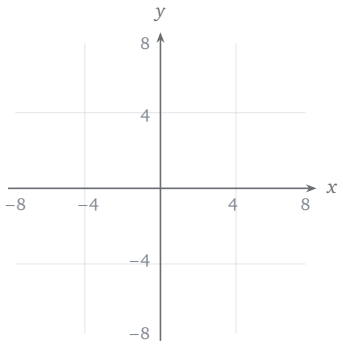
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Textbook · Ex. 12 ●○○

4 Graph $x + y = 4$ by using the intercepts.

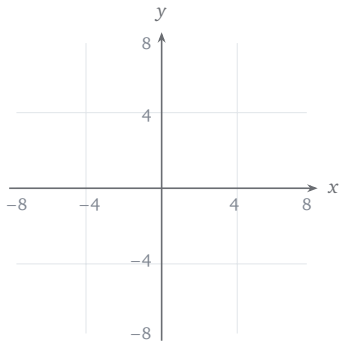
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Textbook · Ex. 13 ●○○

5 Graph $x - y = -3$ by using the intercepts.

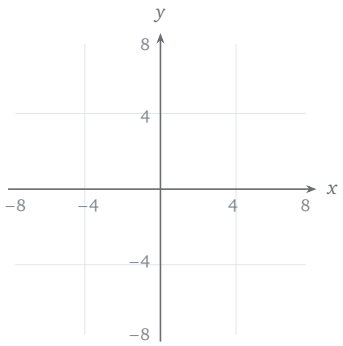
WORKING



Textbook · Ex. 14 ●●○

6 Graph $y = 8 - 6x$ by using the intercepts.

WORKING

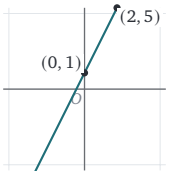


QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q6	MCQ Multiple choice	Find the slope of a line	Page 227 · Exercises 50–52

Multiple choice · Find the slope of the line, then choose the correct answer.

Textbook · Ex. 50

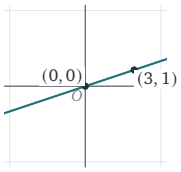
1



(A) $m = 2$ (B) $m = \frac{1}{2}$ (C) $m = -2$ (D) $m = 4$

Textbook · Ex. 51

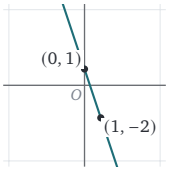
2



(A) $m = 3$ (B) $m = \frac{1}{3}$ (C) $m = -\frac{1}{3}$ (D) $m = 1$

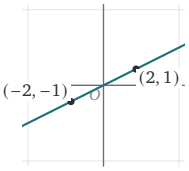
Textbook · Ex. 52

3



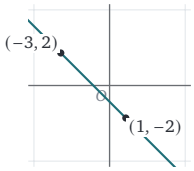
(A) $m = 3$ (B) $m = -\frac{1}{3}$ (C) $m = -3$ (D) $m = -1$

4



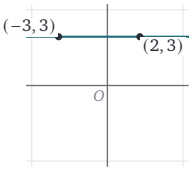
(A) $m = 2$ (B) $m = -\frac{1}{2}$ (C) $m = 1$ (D) $m = \frac{1}{2}$

5



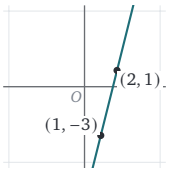
(A) $m = -1$ (B) $m = 1$ (C) $m = -4$ (D) $m = 0$

6



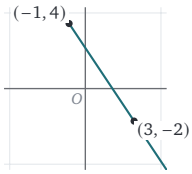
(A) undefined (B) $m = 0$ (C) $m = 1$ (D) $m = 3$

7



(A) $m = \frac{1}{4}$ (B) $m = -4$ (C) $m = 4$ (D) $m = -2$

8



(A) $m = -\frac{2}{3}$ (B) $m = \frac{3}{2}$ (C) $m = -6$ (D) $m = -\frac{3}{2}$

9 $(2, 3),\ (5, 9)$

(A) $m = 2$ (B) $m = \frac{1}{2}$ (C) $m = -2$ (D) $m = 6$

10 $(-1, 4),\ (3, -4)$

(A) $m = 2$ (B) $m = -2$ (C) $m = -\frac{1}{2}$ (D) $m = 0$

11 $(0, -2),\ (6, 1)$

(A) $m = 2$ (B) $m = -\frac{1}{2}$ (C) $m = \frac{1}{2}$ (D) $m = \frac{1}{6}$

12 $(-3, 5),\ (2, 5)$

(A) undefined (B) $m = 1$ (C) $m = -2$ (D) $m = 0$

13 $(4, 1),\ (4, 7)$

(A) undefined (B) $m = 0$ (C) $m = 6$ (D) $m = \frac{1}{6}$

14 $(-6, 1),\ (-2, -5)$

(A) $m = \frac{3}{2}$ (B) $m = -\frac{3}{2}$ (C) $m = -\frac{2}{3}$ (D) $m = -1$

15 $(-5, -2),\ (5, 2)$

(A) $m = \frac{5}{2}$ (B) $m = -\frac{2}{5}$ (C) $m = \frac{2}{5}$ (D) $m = 0$

16 $(2, -4),\ (6, 8)$

(A) $m = \frac{1}{3}$ (B) $m = 1$ (C) $m = -3$ (D) $m = 3$

EXAM FOCUS

LESSON 4-3 Slope-Intercept Form

QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q7	MCQ Multiple choice	Rewrite linear equations in slope-intercept form	Page 235 · Exercises 1–8

Multiple choice · Choose the equation in slope-intercept form, $y = mx + b$.

Textbook · Ex. 1 ●○○

1 Slope 5, y-intercept -3.

(A) $y = 5x - 3$ (B) $y = -3x + 5$

(C) $y = 5x + 3$ (D) $y = 3x - 5$

Textbook · Ex. 2 ●○○

2 Slope -2, y-intercept 7.

(A) $y = 7x - 2$ (B) $y = -2x + 7$

(C) $y = 2x + 7$ (D) $y = -2x - 7$

Textbook · Ex. 3 ●○○

3 Slope -6, y-intercept -2.

(A) $y = -2x - 6$ (B) $y = 6x - 2$

(C) $y = -6x - 2$ (D) $y = -6x + 2$

Textbook · Ex. 4 ●○○

4 Slope 7, y-intercept 1.

(A) $y = x + 7$ (B) $y = 7x - 1$

(C) $y = 7x$ (D) $y = 7x + 1$

Textbook · Ex. 5 ●○○

5 Slope 3, y-intercept 2.

(A) $y = 3x + 2$ (B) $y = 2x + 3$

(C) $y = 3x - 2$ (D) $y = 5x$

Textbook · Ex. 6 ●○○

6 Slope -4, y-intercept -9.

(A) $y = -9x - 4$ (B) $y = -4x - 9$

(C) $y = 4x - 9$ (D) $y = -4x + 9$

Textbook · Ex. 7 ●●○

7 Slope 1, y-intercept -12.

(A) $y = -12x + 1$ (B) $y = x + 12$

(C) $y = x - 12$ (D) $y = -12x$

Textbook · Ex. 8 ●●○

8 Slope 0, y-intercept 8.

(A) $y = 8x$ (B) $x = 8$

(C) $y = 0$ (D) $y = 8$

●○○

9 Rewrite $3x + y = 7$.

(A) $y = -3x + 7$ (B) $y = 3x + 7$

(C) $y = 7x - 3$ (D) $y = -3x - 7$

●●○

10 Rewrite $6x + 2y = 10$.

(A) $y = -6x + 10$ (B) $y = -3x + 5$

(C) $y = 3x + 5$ (D) $y = -3x + 10$

●●○

11 Rewrite $4x - 2y = 8$.

(A) $y = -2x + 4$ (B) $y = 2x + 4$

(C) $y = 2x - 4$ (D) $y = 4x - 8$

●●●

12 Rewrite $x + 3y = 9$.

(A) $y = -3x + 9$ (B) $y = -\frac{1}{3}x + 3$

(C) $y = \frac{1}{3}x + 3$ (D) $y = -x + 3$

●●●

13 Rewrite $-8x + 4y = -12$.

(A) $y = -2x - 3$ (B) $y = 2x + 3$

(C) $y = 2x - 3$ (D) $y = -8x - 12$

●●●

14 Rewrite $3y - 2x = 12$.

(A) $y = 2x + 12$ (B) $y = -\frac{2}{3}x + 4$

(C) $y = \frac{3}{2}x + 4$ (D) $y = \frac{2}{3}x + 4$

EXAM FOCUS

LESSON 4-4 Transformations of Linear Functions

QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q8	MCQ Multiple choice	Apply dilations to linear functions	Page 248 · Exercises 10–15

Multiple choice · Describe the dilation of $g(x)$ compared with the parent function $f(x) = x$.

Textbook · Ex. 10 ●○○

1 $g(x) = 5(x)$

A vertical stretch, factor 5 (steeper)
B vertical compression, factor $\frac{1}{5}$ (less steep)
C vertical compression, factor 5 (less steep)
D horizontal stretch, factor 5 (less steep)

Textbook · Ex. 11 ●○○

2 $g(x) = \frac{1}{3}(x)$

A vertical stretch, factor 3 (steeper)
B vertical compression, factor $\frac{1}{3}$ (less steep)
C vertical compression, factor 3 (less steep)
D horizontal compression, factor $\frac{1}{3}$ (steeper)

Textbook · Ex. 12 ●○○

3 $g(x) = 1.5(x)$

A vertical compression, factor 1.5 (less steep)
B vertical stretch, factor $\frac{2}{3}$ (steeper)
C vertical stretch, factor 1.5 (steeper)
D horizontal stretch, factor 1.5 (less steep)

Textbook · Ex. 13 ●●○

4 $g(x) = (3x)$

A vertical compression, factor $\frac{1}{3}$ (less steep)
B horizontal compression, factor 3 (steeper)
C horizontal stretch, factor 3 (less steep)
D horizontal compression, factor $\frac{1}{3}$ (steeper)

Textbook · Ex. 14 ●●○

5 $g(x) = \left(\frac{3}{4}x\right)$

A horizontal stretch, factor $\frac{4}{3}$ (less steep)
B horizontal stretch, factor $\frac{3}{4}$ (less steep)
C vertical stretch, factor $\frac{4}{3}$ (steeper)
D horizontal compression, factor $\frac{4}{3}$ (steeper)

Textbook · Ex. 15 ●●○

6 $g(x) = (0.4x)$

A horizontal compression, factor 0.4 (steeper)
B horizontal stretch, factor 2.5 (less steep)
C vertical stretch, factor 2.5 (steeper)
D horizontal stretch, factor 0.4 (less steep)

●○○

7 $g(x) = 4(x)$

A horizontal stretch, factor 4 (less steep)
B vertical compression, factor $\frac{1}{4}$ (less steep)
C vertical stretch, factor 4 (steeper)
D vertical compression, factor 4 (less steep)

●○○

8 $g(x) = \frac{1}{2}(x)$

A vertical stretch, factor 2 (steeper)
B vertical stretch, factor $\frac{1}{2}$ (steeper)
C vertical compression, factor 2 (less steep)
D vertical compression, factor $\frac{1}{2}$ (less steep)

●●○

9 $g(x) = (5x)$

A horizontal compression, factor $\frac{1}{5}$ (steeper)
B horizontal compression, factor 5 (steeper)
C vertical compression, factor $\frac{1}{5}$ (less steep)
D horizontal stretch, factor 5 (less steep)

●●○

10 $g(x) = \left(\frac{1}{5}x\right)$

A horizontal compression, factor $\frac{1}{5}$ (steeper)
B horizontal stretch, factor 5 (less steep)
C vertical stretch, factor 5 (steeper)
D horizontal stretch, factor $\frac{1}{5}$ (less steep)

EXAM FOCUS

LESSON 4-5 Arithmetic Sequences

QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q24	FRQ Free response	Apply the arithmetic sequence formula	**Page 256** · Exercises 23–26

Written answer · Write the function, graph it, then answer part c. Show your steps.

Textbook · Ex. 23 ●●○

1 **Water.** Each cup holds 4 oz. Wanda starts with a 128-oz cooler.
a. Write a function. **b.** Graph it. **c.** How much water is left after the 14th cup?

SOLUTION

Final answer

Textbook · Ex. 24 ●○○

2 **Theater.** 20 seats in row 1, 22 in row 2, 24 in row 3, and so on for 25 rows.
a. Write a function. **b.** Graph it. **c.** How many seats are in the last row?

SOLUTION

Final answer

Textbook · Ex. 25 ●●●

3 **Postage.** $0.88 for the first ounce, $0.17 for each extra ounce: $0.88, $1.05, $1.22, …
a. Write a function. **b.** Graph it. **c.** How much did an envelope weigh that cost $2.07?

SOLUTION

Final answer

Textbook · Ex. 26 ●○○

4 **Downloads.** 1 episode costs $1.99, 2 cost $3.98, 3 cost $5.97.
a. Write a function. **b.** Graph it. **c.** What is the cost of 9 episodes?

SOLUTION

Final answer

●●○

5 **Savings.** Omar has $50 in week 1 and adds $15 every week: $50, $65, $80, …
a. Write a function. **b.** Graph it. **c.** How much does he have in week 10?

SOLUTION

Final answer

●●●

6 **Training.** Sara runs 12 min on day 1, then 3 min more each day: 12, 15, 18, …
a. Write a function. **b.** Graph it. **c.** On which day does she run 60 min?

SOLUTION

Final answer

EXAM FOCUS

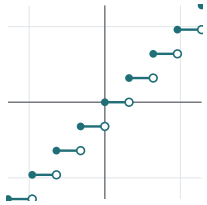
LESSON 4-6 Piecewise and Step Functions

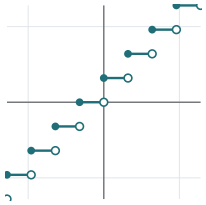
QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q9	MCQ Multiple choice	Identify and graph step functions	**Page 261** · Example 2

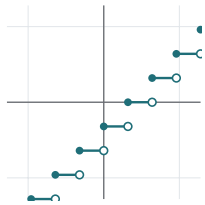
Multiple choice · Use the greatest integer function $[\![x]\!]$ (round down).

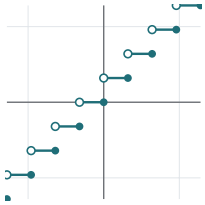
Textbook · Ex. 2 ●●○

1 Which graph shows $f(x) = [\![x + 1]\!]$?

A 

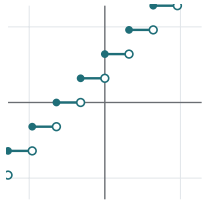
B 

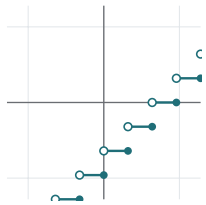
C 

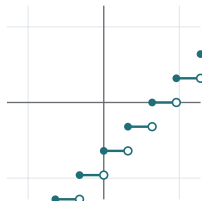
D 

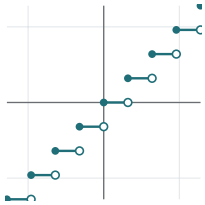
●●○

2 Which graph shows $f(x) = [\![x - 2]\!]$?

A 

B 

C 

D 

Textbook · Ex. 2 ●○○

3 $[\![x + 1]\!]$ at $x = -1.75$

A -1 B 0 C -2 D -0.75

●○○

4 $[\![x - 2]\!]$ at $x = 0.25$

A -1.75 B -1 C 0 D -2

Textbook · Ex. 2 ●●○

5 What is the **domain** of $f(x) = [\![x + 1]\!]$?

A all integers B $x \geq -1$ C all real numbers D whole numbers

Textbook · Ex. 2 ●●○

6 What is the **range** of $f(x) = [\![x + 1]\!]$?

A all real numbers B all integers C $y \geq 1$ D $y \geq 0$

●●○

7 How does the graph of $[\![x + 3]\!]$ compare with the graph of $[\![x]\!]$?

A 3 units left B 3 units right C 3 units down D 3 times as steep

●●●

8 For $f(x) = [\![x]\!]$, which point is **not** on the graph?

A $(0.5, 0)$ B $(-0.5, -1)$ C $(2, 2)$ D $(1, 0)$

EXAM FOCUS

LESSON 4-6 **Piecewise and Step Functions**

QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q10	MCQ Multiple choice	Identify and graph step functions	**Page 265** · Exercises 18a–d

Multiple choice · Evaluate each step function. Inside first, round down, then multiply.

Textbook · Ex. 18a ●○○

1 $f(x) = 2[\![x - 1]\!]$, $f(1.5) = ?$

A 0 B 1 C 2 D −1

Textbook · Ex. 18b ●○○

2 $f(x) = 2[\![x - 1]\!]$, $f(2.2) = ?$

A 2.4 B 2 C 4 D 1

Textbook · Ex. 18c ●●○

3 $f(x) = 2[\![x - 1]\!]$, $f(9.7) = ?$

A 18 B 17.4 C 16 D 8

Textbook · Ex. 18d ●●●

4 $f(x) = 2[\![x - 1]\!]$, $f(-1.25) = ?$

A −4 B −4.5 C −2 D −6

●○○

5 $g(x) = 3[\![x + 2]\!]$, $g(0.5) = ?$

A 6 B 7.5 C 9 D 2

●●○

6 $g(x) = 3[\![x + 2]\!]$, $g(-0.5) = ?$

A 6 B 3 C 4.5 D 0

●●●

7 $g(x) = 3[\![x + 2]\!]$, $g(-2.5) = ?$

A 0 B −1.5 C −3 D −6

●●○

8 $h(x) = [\![x]\!] + 4$, $h(-1.2) = ?$

A 3 B 2.8 C −5 D 2

●●○

9 $k(x) = -2[\![x]\!]$, $k(3.6) = ?$

A −6 B −8 C 6 D −7.2

●●○

10 $m(x) = [\![2x]\!]$, $m(1.7) = ?$

A 2 B 3 C 4 D 3.4

●●●

11 $q(x) = 4[\![x - 0.5]\!]$, $q(3) = ?$

A 10 B 12 C 8 D 2

●●●

12 $r(x) = [\![\frac{x}{2}]\!]$, $r(-3) = ?$

A −1.5 B −1 C 1 D −2

EXAM FOCUS

LESSON 4-7 **Absolute Value Functions**

QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q11	MCQ Multiple choice	Apply translations to absolute value functions	**Page 267** · Example 1

Multiple choice · Describe the translation of $g(x)$ compared with $f(x) = |x|$.

Textbook · Ex. 1 ●○○

1 $g(x) = |x| - 3$

(A) 3 units down (B) 3 units up (C) 3 units left (D) 3 units right

●○○

2 $g(x) = |x| + 4$

(A) 4 units down (B) 4 units up (C) 4 units right (D) 4 units left

●○○

3 $g(x) = |x - 3|$

(A) 3 units left (B) 3 units down (C) 3 units right (D) 3 units up

●○○

4 $g(x) = |x + 6|$

(A) 6 units right (B) 6 units up (C) 6 units down (D) 6 units left

●●○

5 $g(x) = |x| - 7$

(A) 7 units down (B) 7 units left (C) 7 units up (D) 7 units right

●●○

6 $g(x) = |x + 1|$

(A) 1 unit right (B) 1 unit left (C) 1 unit up (D) 1 unit down

●●○

7 Describe $g(x) = |x - 2| + 5$ compared with $f(x) = |x|$.

(A) 2 left, 5 up (B) 2 right, 5 down (C) 2 right, 5 up (D) 5 right, 2 up

●●●

8 Describe $g(x) = |x + 4| - 1$ compared with $f(x) = |x|$.

(A) 4 right, 1 down (B) 4 left, 1 up (C) 1 left, 4 down (D) 4 left, 1 down

●●●

9 Which function is $f(x) = |x|$ translated **5 units right and 2 units down**?

(A) $g(x) = |x - 5| - 2$ (B) $g(x) = |x + 5| - 2$ (C) $g(x) = |x - 2| + 5$ (D) $g(x) = |x - 5| + 2$

●●●

10 What is the vertex of $g(x) = |x + 3| + 6$?

(A) $(3, 6)$ (B) $(-3, 6)$ (C) $(6, -3)$ (D) $(-3, -6)$

EXAM FOCUS

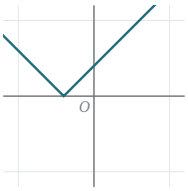
LESSON 4-7 **Absolute Value Functions**

QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q12	MCQ Multiple choice	Graph absolute value functions	**Page 277** · Exercises 7–12

Multiple choice · Use the graph to choose the equation of the function.

Textbook · Ex. 7 ●○○

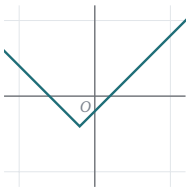
1



A $|x+2|$ B $|x-2|$ C $|x|+2$ D $|x|-2$

Textbook · Ex. 8 ●●○

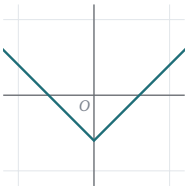
2



A $|x-1|-2$ B $|x+1|-2$ C $|x+2|-1$ D $|x+1|+2$

Textbook · Ex. 9 ●○○

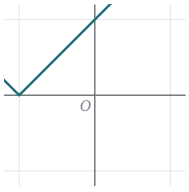
3



A $|x-3|$ B $|x|+3$ C $|x|-3$ D $|x+3|$

Textbook · Ex. 10 ●●○

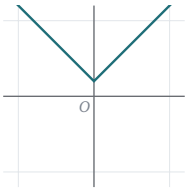
4



A $|x-5|$ B $|x|-5$ C $|x|+5$ D $|x+5|$

Textbook · Ex. 11 ●○○

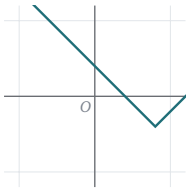
5



A $|x|+1$ B $|x|-1$ C $|x+1|$ D $|x-1|$

Textbook · Ex. 12 ●●●

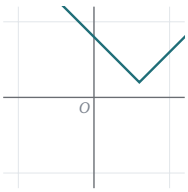
6



A $|x+4|-2$ B $|x-4|-2$ C $|x-2|-4$ D $|x-4|+2$

●●○

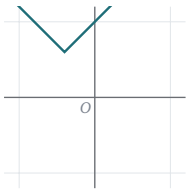
7



A $|x+3|+1$ B $|x-1|+3$ C $|x-3|+1$ D $|x-3|-1$

●●○

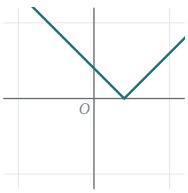
8



A $|x-2|+3$ B $|x+3|-2$ C $|x+2|-3$ D $|x+2|+3$

●●○

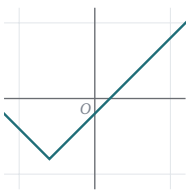
9



A $|x-2|$ B $|x+2|$ C $|x|-2$ D $|x|+2$

●●●

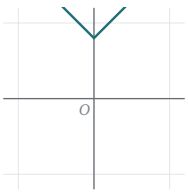
10



A $|x-3|-4$ B $|x+3|-4$ C $|x+4|-3$ D $|x+3|+4$

●●○

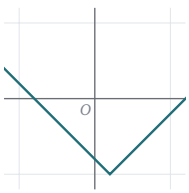
11



A $|x+4|$ B $|x|-4$ C $|x|+4$ D $|x-4|$

●●●

12



A $|x+1|-5$ B $|x-5|+1$ C $|x-1|+5$ D $|x-1|-5$

EXAM FOCUS

LESSON 5-1 Writing Equations in Slope-Intercept Form

QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q13	MCQ Multiple choice	Write an equation of a line in slope-intercept form given the slope and one point	**Page 293** · Exercises 33–42

Multiple choice · Choose the correct answer.

Textbook · Ex. 33 ●○○

1 Is $(3, -1)$ on $y = \frac{1}{3}x + 5$?

A Yes: $y = -1$

B No: $y = 6$

C No: $y = 4$

D Yes: $y = 6$

Textbook · Ex. 34 ●○○

2 Is $(6, -2)$ on $y = \frac{1}{2}x - 5$?

A Yes: $y = -2$

B No: $y = -8$

C No: $y = 2$

D Yes: $y = 8$

Textbook · Ex. 35 ●●○

3 Is $(15, -13)$ on $y = -\frac{1}{5}x - 10$?

A No: $y = -7$

B No: $y = -25$

C Yes: $y = -13$

D No: $y = 13$

Textbook · Ex. 36 ●●○

4 Is $(3, 3)$ on $y = -\frac{2}{3}x + 1$?

A Yes: $y = 3$

B No: $y = 3$

C Yes: $y = -1$

D No: $y = -1$

Textbook · Ex. 37 ●○○

5 Another point on the line through $(2, -4)$, $(4, -2)$?

A $(6,\ 0)$

B $(6,\ -6)$

C $(0,\ -4)$

D $(3,\ -4)$

Textbook · Ex. 38 ●○○

6 Another point on the line through $(0, 5)$, $(4, 1)$?

A $(8,\ 5)$

B $(2,\ 3)$

C $(1,\ 1)$

D $(5,\ 4)$

Textbook · Ex. 39 ●●○

7 Another point on the line through $(-3, 1)$, $(-1, -3)$?

A $(1,\ -1)$

B $(0,\ -4)$

C $(1,\ -7)$

D $(-5,\ -3)$

Textbook · Ex. 40 ●●○

8 Another point on the line through $(0, 4)$, $(2, 5)$?

A $(4,\ 8)$

B $(1,\ 6)$

C $(-2,\ 5)$

D $(4,\ 6)$

Textbook · Ex. 41 ●●●

9 Another point on the line through $(-2, 9)$, $(2, -1)$?

A $(6,\ -11)$

B $(6,\ 9)$

C $(4,\ -3)$

D $(0,\ 5)$

Textbook · Ex. 42 ●●●

10 Another point on the line through $(3, 0)$, $(12, 3)$?

A $(6,\ 3)$

B $(6,\ 1)$

C $(9,\ 1)$

D $(0,\ 3)$

●○○

11 Slope 3, through $(1, 5)$.

A $y = 3x + 5$

B $y = 3x - 2$

C $y = 3x + 2$

D $y = 5x + 3$

●○○

12 Slope -2, through $(2, 1)$.

A $y = -2x + 1$

B $y = -2x - 3$

C $y = 2x - 3$

D $y = -2x + 5$

EXAM FOCUS

LESSON 5-2 Writing Equations in Standard and Point-Slope Forms

QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q14	MCQ Multiple choice	Write equations of lines in point-slope form given two points	**Page 297, 303** · Example 2 · Exercises 4–7

Multiple choice · Choose an equation in point-slope form for the line through the two points.

Textbook · Ex. 2 ●○○

1 $(2, -7)$ and $(6, -3)$

A $y + 7 = x - 2$
B $y - 7 = x + 2$
C $y + 7 = 2(x - 2)$
D $y + 7 = -(x - 2)$

Textbook · Ex. 4 ●●○

2 $(-4, 6)$ and $(-2, 22)$

A $y - 6 = 8(x - 4)$
B $y - 6 = 8(x + 4)$
C $y + 6 = 8(x + 4)$
D $y - 6 = \frac{1}{8}(x + 4)$

Textbook · Ex. 5 ●●○

3 $(1, -3)$ and $(4, -15)$

A $y - 3 = -4(x - 1)$
B $y + 3 = 4(x - 1)$
C $y + 3 = -4(x - 1)$
D $y + 3 = -4(x + 1)$

Textbook · Ex. 6 ●○○

4 $(4, -6)$ and $(6, -4)$

A $y - 6 = x - 4$
B $y + 6 = -(x - 4)$
C $y + 6 = x + 4$
D $y + 6 = x - 4$

Textbook · Ex. 7 ●●○

5 $(3, 3)$ and $(6, 7)$

A $y - 3 = \frac{4}{3}(x - 3)$
B $y - 3 = \frac{3}{4}(x - 3)$
C $y + 3 = \frac{4}{3}(x + 3)$
D $y - 3 = 4(x - 3)$

●○○

6 $(1, 2)$ and $(3, 8)$

A $y - 2 = 2(x - 1)$
B $y - 2 = 3(x - 1)$
C $y + 2 = 3(x + 1)$
D $y - 1 = 3(x - 2)$

●●○

7 $(-2, 5)$ and $(2, -3)$

A $y - 5 = 2(x + 2)$
B $y + 5 = -2(x - 2)$
C $y - 5 = -2(x + 2)$
D $y - 5 = -\frac{1}{2}(x + 2)$

●●○

8 $(0, -1)$ and $(5, 9)$

A $y - 1 = 2x$
B $y + 1 = \frac{1}{2}x$
C $y + 1 = 2(x + 5)$
D $y + 1 = 2x$

●●●

9 $(-6, 4)$ and $(3, -2)$

A $y - 4 = -\frac{2}{3}(x + 6)$
B $y - 4 = -\frac{3}{2}(x + 6)$
C $y + 4 = -\frac{2}{3}(x - 6)$
D $y - 4 = \frac{2}{3}(x + 6)$

●●●

10 $(-5, -1)$ and $(-1, 7)$

A $y - 1 = 2(x - 5)$
B $y + 1 = 2(x + 5)$
C $y + 1 = \frac{1}{2}(x + 5)$
D $y + 1 = -2(x + 5)$

EXAM FOCUS

LESSON 5-2 **Writing Equations in Standard and Point-Slope Forms**

QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q15	MCQ Multiple choice	Write an equation of the line that passes through a given point, parallel to a given line	**Page 304** · Exercises 25–29

Multiple choice · Choose the line in slope-intercept form through the point, parallel to the given line.

Textbook · Ex. 25 ●○○

1 $(3, -2)$; parallel to $y = x + 4$

A $y = x - 5$ B $y = x + 4$

C $y = -x + 1$ D $y = x + 1$

Textbook · Ex. 26 ●○○

2 $(4, -3)$; parallel to $y = 3x - 5$

A $y = 3x - 5$ B $y = 3x - 15$

C $y = -\frac{1}{3}x - \frac{5}{3}$ D $y = 3x + 9$

Textbook · Ex. 27 ●○○

3 $(0, 2)$; parallel to $y = -5x + 8$

A $y = \frac{1}{5}x + 2$ B $y = -5x + 8$

C $y = -5x + 2$ D $y = 2x - 5$

Textbook · Ex. 28 ●●○

4 $(-4, 2)$; parallel to $y = -\frac{1}{2}x + 6$

A $y = 2x + 10$ B $y = -\frac{1}{2}x + 4$

C $y = -\frac{1}{2}x + 6$ D $y = -\frac{1}{2}x$

Textbook · Ex. 29 ●●●

5 $(-2, 3)$; parallel to $y = -\frac{3}{4}x + 4$

A $y = -\frac{3}{4}x + \frac{3}{2}$ B $y = -\frac{3}{4}x + \frac{9}{2}$

C $y = \frac{4}{3}x + \frac{17}{3}$ D $y = -\frac{3}{4}x + 4$

●○○

6 $(1, 6)$; parallel to $y = 2x - 7$

A $y = 2x - 7$ B $y = 2x + 4$

C $y = -\frac{1}{2}x + \frac{13}{2}$ D $y = 2x + 6$

●●○

7 $(-3, -1)$; parallel to $y = -4x + 2$

A $y = -4x - 1$ B $y = 4x + 11$

C $y = -4x - 13$ D $y = \frac{1}{4}x - \frac{1}{4}$

●●○

8 $(6, 0)$; parallel to $y = \frac{1}{3}x + 5$

A $y = \frac{1}{3}x + 5$ B $y = -3x + 18$

C $y = \frac{1}{3}x + 2$ D $y = \frac{1}{3}x - 2$

●●●

9 $(2, -5)$; parallel to $2x + y = 7$

A $y = -2x - 1$ B $y = 2x - 9$

C $y = -2x + 7$ D $y = \frac{1}{2}x - 6$

●●●

10 $(-4, 1)$; parallel to $x - 2y = 6$

A $y = -2x - 7$ B $y = \frac{1}{2}x + 3$

C $y = \frac{1}{2}x - 3$ D $y = -\frac{1}{2}x - 1$

EXAM FOCUS

LESSON 5-2 Writing Equations in Standard and Point-Slope Forms

QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q16	MCQ Multiple choice	Create and identify equations of parallel or perpendicular lines	**Page 304, 305** · Exercises 31–43

Multiple choice · Decide how the lines are related, or rewrite the equation.

Textbook · Ex. 31 ●○○

1 $y = 4x + 3$ and $4x + y = 3$

A parallel B perpendicular C neither D same line

Textbook · Ex. 32 ●○○

2 $y = -2x$ and $2x + y = 3$

A parallel B perpendicular C neither D same line

Textbook · Ex. 33 ●●○

3 $3x + 5y = 10$ and $5x - 3y = -6$

A parallel B perpendicular C neither D same line

Textbook · Ex. 34 ●●○

4 $-3x + 4y = 8$ and $-4x + 3y = -6$

A parallel B perpendicular C neither D same line

Textbook · Ex. 35 ●●○

5 $2x + 5y = 15$ and $3x + 5y = 15$

A parallel B perpendicular C neither D same line

Textbook · Ex. 36 ●●○

6 $2x + 7y = -35$ and $4x + 14y = -42$

A parallel B perpendicular C neither D same line

Textbook · Ex. 37 ●●●

7 Standard form: x-int. 4, y-int. 5

A $4x + 5y = 20$ B $5x - 4y = 20$ C $x + y = 9$ D $5x + 4y = 20$

Textbook · Ex. 38 ●●○

8 $y = mx + b$ for $y + 3 = -\frac{1}{3}(2x + 6)$:

A $y = -\frac{2}{3}x - 1$ B $y = -\frac{2}{3}x - 5$ C $y = -\frac{2}{3}x + 5$ D $y = -2x - 9$

Textbook · Ex. 39 ●○○

9 $y = mx + b$ for $y + 4 = 3(3x + 3)$:

A $y = 9x + 5$ B $y = 9x + 13$ C $y = 3x + 5$ D $y = 9x - 1$

Textbook · Ex. 40 ●○○

10 $y = mx + b$ for $y - 6 = -3(x + 2)$:

A $y = -3x + 12$ B $y = -3x + 4$ C $y = -3x$ D $y = 3x$

Textbook · Ex. 41 ●●○

11 $y = mx + b$ for $y - 9 = -6(x + 9)$:

A $y = -6x + 63$ B $y = -6x - 63$ C $y = -6x + 45$ D $y = -6x - 45$

Textbook · Ex. 42 ●●●

12 $y = mx + b$ for $y + 4 = \frac{2}{3}(x + 7)$:

A $y = \frac{2}{3}x + 3$ B $y = \frac{2}{3}x + \frac{2}{3}$ C $y = \frac{2}{3}x - \frac{26}{3}$ D $y = \frac{3}{2}x + \frac{13}{2}$

Textbook · Ex. 43 ●●●

13 $Ax + By = C$ for $y + 7 = \frac{9}{10}(x + 3)$:

A $9x - 10y = -43$ B $9x + 10y = 43$ C $9x - 10y = 43$ D $10x - 9y = 43$

EXAM FOCUS

LESSON 5-2 Writing Equations in Standard and Point-Slope Forms

QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q25	FRQ Free response	Create and identify equations of parallel or perpendicular lines	**Page 301, 302** · Examples 8–9

Written answer · Show every step. Write the final equation in slope-intercept form.

Textbook · Ex. 8 ●●○

1 Write an equation of the line through $(1, -2)$ that is **perpendicular** to $3x + 2y = 12$.

SOLUTION

Final answer

Textbook · p. 302 ●●○

2 Write an equation of the line through $(5, 0)$ that is **perpendicular** to $x - 6y = 1$.

SOLUTION

Final answer

Textbook · p. 301 ●○○

3 Write an equation of the line through $(8, 2)$ that is **parallel** to $y = \frac{3}{4}x + 2$.

SOLUTION

Final answer

Textbook · Ex. 9 ●●○

4 Are $\overleftrightarrow{AB}$ and $\overleftrightarrow{EF}$ parallel, perpendicular, or neither? $A(6, 8)$, $B(2, 5)$, $E(-6, -3)$, $F(0, 5)$.

SOLUTION

Final answer

Textbook · p. 302 ●●○

5 Are $\overleftrightarrow{CD}$ and $\overleftrightarrow{KL}$ parallel, perpendicular, or neither? $C(4, 10)$, $D(-1, 12)$, $K(6, -5)$, $L(1, -3)$.

SOLUTION

Final answer

●●●

6 Write an equation of the line through $(-3, 1)$ that is **perpendicular** to $y = 3x - 2$.

SOLUTION

Final answer

EXAM FOCUS

LESSON 5-6 **Inverses of Linear Functions**

QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q17	MCQ Multiple choice	Construct the inverses of relations	**Page 331** · Exercises 4–9

Multiple choice · Find the inverse of each relation.

Textbook · Ex. 4 ●○○

1 $\{(-3, 2),\ (-1, 8),\ (1, 14),\ (3, 20)\}$

A $\{(2, -3), (8, -1), (14, 1), (20, 3)\}$

B $\{(3, -2), (1, -8), (-1, -14), (-3, -20)\}$

C $\{(-2, 3), (-8, 1), (-14, -1), (-20, -3)\}$

D $\{(2, 3), (8, 1), (14, -1), (20, -3)\}$

Textbook · Ex. 5 ●○○

2 $\{(5, -3),\ (2, -9),\ (-1, -15),\ (-4, -21)\}$

A $\{(-5, 3), (-2, 9), (1, 15), (4, 21)\}$

B $\{(-3, 5), (-9, 2), (-15, -1), (-21, -4)\}$

C $\{(3, -5), (9, -2), (15, 1), (21, 4)\}$

D $\{(-3, -5), (-9, -2), (-15, 1), (-21, 4)\}$

Textbook · Ex. 6 ●○○

3 $\{(4, 6),\ (3, 1),\ (2, -4),\ (1, -9)\}$

A $\{(-4, -6), (-3, -1), (-2, 4), (-1, 9)\}$

B $\{(6, -4), (1, -3), (-4, -2), (-9, -1)\}$

C $\{(6, 4), (1, 3), (-4, 2), (-9, 1)\}$

D $\{(4, 6), (3, 1), (2, -4), (1, -9)\}$

Textbook · Ex. 7 ●●○

4 $\{(-1, 16),\ (-2, 12),\ (-3, 8),\ (-4, 4)\}$

A $\{(1, -16), (2, -12), (3, -8), (4, -4)\}$

B $\{(16, 1), (12, 2), (8, 3), (4, 4)\}$

C $\{(-16, 1), (-12, 2), (-8, 3), (-4, 4)\}$

D $\{(16, -1), (12, -2), (8, -3), (4, -4)\}$

Textbook · Ex. 8 ●●○

5 $\{(-5, 13),\ (6, 10.8),\ (3, 11.4),\ (-10, 14)\}$

A $\{(13, -5), (10.8, 6), (11.4, 3), (14, -10)\}$

B $\{(5, -13), (-6, -10.8), (-3, -11.4), (10, -14)\}$

C $\{(13, 5), (10.8, -6), (11.4, -3), (14, 10)\}$

D $\{(-13, 5), (-10.8, -6), (-11.4, -3), (-14, 10)\}$

Textbook · Ex. 9 ●●○

6 $\{(-4, -49),\ (8, 35),\ (-1, -28),\ (4, 7)\}$

A $\{(4, 49), (-8, -35), (1, 28), (-4, -7)\}$

B $\{(-49, -4), (35, 8), (-28, -1), (7, 4)\}$

C $\{(49, 4), (-35, -8), (28, 1), (-7, -4)\}$

D $\{(-49, 4), (35, -8), (-28, 1), (7, -4)\}$

●○○

7 The inverse of the point $(7, -2)$ is

A $(-7, 2)$

B $(2, -7)$

C $(-2, 7)$

D $(7, 2)$

●●○

8 A relation has range $\{0, 2, 4\}$. What is the **domain** of its inverse?

A $\{0, -2, -4\}$

B all real numbers

C $\{1, 2, 3\}$

D $\{0, 2, 4\}$

EXAM FOCUS

LESSON 6-1 **Solving One-Step Inequalities**

QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q18	MCQ Multiple choice	Solve linear inequalities by using subtraction	**Page 347** · Exercises 7–12

Multiple choice · Write the inequality shown by the graph, or solve the inequality.

Textbook · Ex. 7 ●○○

1



(A) $x < -1$ (B) $x \leq -1$
(C) $x > -1$ (D) $x \geq -1$

Textbook · Ex. 8 ●○○

2



(A) $x < -6$ (B) $x > -6$
(C) $x \geq -6$ (D) $x \leq -6$

Textbook · Ex. 9 ●○○

3



(A) $x > 5$ (B) $x \leq 5$
(C) $x < 5$ (D) $x \geq 5$

Textbook · Ex. 10 ●○○

4



(A) $x < 1$ (B) $x \geq 1$
(C) $x > 1$ (D) $x \leq 1$

Textbook · Ex. 11 ●●○

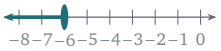
5



(A) $x \geq -5$ (B) $x > -5$
(C) $x \leq -5$ (D) $x < -5$

Textbook · Ex. 12 ●●○

6



(A) $x < -6$ (B) $x \leq -6$
(C) $x \geq -6$ (D) $x > -6$

●○○

7 $m + 4 < 9$

(A) $m < 13$ (B) $m > 5$
(C) $m < 5$ (D) $m \leq 5$

●○○

8 $x + 7 \geq 3$

(A) $x \geq 10$ (B) $x \leq -4$
(C) $x \geq 4$ (D) $x \geq -4$

●●○

9 $13 > 18 + r$

(A) $r < -5$ (B) $r > -5$
(C) $r < 5$ (D) $r < 31$

●●○

10 $5 + c \leq 1$

(A) $c \leq 6$ (B) $c \leq -4$
(C) $c \geq -4$ (D) $c \leq 4$

●●○

11 $11 + m \geq 15$

(A) $m \geq 26$ (B) $m \leq 4$
(C) $m \geq 4$ (D) $m > 4$

●●●

12 $-7 > 20 + c$

(A) $c > -27$ (B) $c < 13$
(C) $c < 27$ (D) $c < -27$

●●●

13 $2a \leq -4 + a$

(A) $a \leq -4$ (B) $a \geq -4$
(C) $a \leq 4$ (D) $a \leq -\frac{4}{3}$

●●●

14 $3y \leq 2y - 6$

(A) $y \geq -6$ (B) $y \leq -6$
(C) $y \leq 6$ (D) $y \leq -\frac{6}{5}$

EXAM FOCUS

LESSON 6-2 Solving Multi-Step Inequalities

QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q19	MCQ Multiple choice	Solve linear inequalities involving more than one operation	**Page 354** · Examples 2–3

Multiple choice · Write the inequality if needed, then solve it.

Textbook · Ex. 6 ●○○

1 Five times a number minus one is greater than or equal to negative eleven.

(A) $n \geq -2$ (B) $n \leq -2$

(C) $n \geq -\frac{12}{5}$ (D) $n \geq 2$

Textbook · Ex. 7 ●○○

2 Twenty-one is greater than the sum of fifteen and two times a number.

(A) $n > 3$ (B) $n < 3$

(C) $n < 18$ (D) $n < \frac{3}{2}$

Textbook · Ex. 9 ●●○

3 A number divided by eight minus thirteen is greater than negative six.

(A) $n > -56$ (B) $n > \frac{7}{8}$

(C) $n > 56$ (D) $n < 56$

Textbook · Ex. 10 ●●○

4 The sum of the opposite of a number and six is less than or equal to five.

(A) $n \leq 1$ (B) $n \leq -1$

(C) $n \geq 11$ (D) $n \geq 1$

Textbook · Ex. 11 ●●●

5 Thirty-seven is less than the difference of seven and ten times a number.

(A) $n < -3$ (B) $n > -3$

(C) $n < 3$ (D) $n > 3$

Textbook · Ex. 12 ●●●

6 Eight minus a number divided by three is greater than or equal to eleven.

(A) $n \geq -9$ (B) $n \leq -9$

(C) $n \leq 9$ (D) $n \geq -1$

Textbook · Ex. 14 ●●●

7 The difference of three times a number and six is greater than or equal to the sum of fifteen and twenty-four times a number.

(A) $n \geq -1$ (B) $n \leq 1$

(C) $n \leq -1$ (D) $n \geq 1$

Textbook · Ex. 15 ●●●

8 The sum of fifteen times a number and thirty is less than the difference of ten times a number and forty-five.

(A) $n > -15$ (B) $n < -3$

(C) $n < 15$ (D) $n < -15$

Textbook · Ex. 16 ●●○

9 $-3(7n + 3) < 6n$

(A) $n > -\frac{1}{3}$ (B) $n < -\frac{1}{3}$

(C) $n > \frac{1}{3}$ (D) $n > -3$

Textbook · Ex. 17 ●●○

10 $21 \geq 3(a - 7) + 9$

(A) $a \geq 11$ (B) $a \leq 11$

(C) $a \leq 7$ (D) $a \leq \frac{11}{3}$

Textbook · Ex. 18 ●●●

11 $2y + 4 > 2(3 + y)$

(A) $y > 1$ (B) all reals

(C) $\varnothing$ (D) $y > 0$

Textbook · Ex. 19 ●●●

12 $3(2 - b) < 10 - 3(b - 6)$

(A) $\varnothing$ (B) $b < 4$

(C) $b > -4$ (D) all reals

Textbook · Ex. 20 ●●○

13 $7 + t \leq 2(t + 3) + 2$

(A) $t \geq -1$ (B) $t \leq -1$

(C) $t \geq 1$ (D) $t \geq -3$

Textbook · Ex. 21 ●●●

14 $8a + 2(1 - 5a) \leq 20$

(A) $a \leq -9$ (B) $a \geq -9$

(C) $a \geq -11$ (D) $a \leq 9$

EXAM FOCUS

LESSON 6-3 **Solving Compound Inequalities**

QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q21	FRQ Free response	Solve and graph linear inequalities containing the word and	**Page 363** · Exercises 1, 2, 5, 6, 7, 10, 11, 12

Written answer · Solve each compound inequality, then graph the solution set on the number line.

Textbook · Ex. 1 ●○○

1 Solve $f - 6 < 5$ and $f - 4 \geq 2$.



SOLUTION

Final answer

Textbook · Ex. 2 ●○○

2 Solve $n + 2 \leq -5$ and $n + 6 \geq -6$.



SOLUTION

Final answer

Textbook · Ex. 5 ●●○

3 Solve $-5 < 3p + 7 \leq 22$.



SOLUTION

Final answer

Textbook · Ex. 6 ●●○

4 Solve $-3 \leq 7c + 4 < 18$.



SOLUTION

Final answer

Textbook · Ex. 7 ●●○

5 Solve $5h - 4 \geq 6$ and $7h + 11 < 32$.



SOLUTION

Final answer

Textbook · Ex. 10 ●●●

6 Solve $-4a + 13 \geq 29$ and $10 < 6a - 14$.



SOLUTION

Final answer

Textbook · Ex. 11 ●●●

7 Solve $3b + 2 < 5b - 6 \leq 2b + 9$.



SOLUTION

Final answer

Textbook · Ex. 12 ●●●

8 Solve $-2a + 3 \geq 6a - 1 > 3a - 10$.



SOLUTION

Final answer

EXAM FOCUS

LESSON 6-4 **Solving Absolute Value Inequalities**

QUESTION	TYPE	LEARNING OUTCOME	TEXTBOOK
Q20	MCQ Multiple choice	Solve and graph absolute value inequalities (<)	**Page 371** · Exercises 1–10

Multiple choice · Solve each inequality and choose the solution set.

Textbook · Ex. 1 ●○○

1 $|x + 8| < 16$

(A) $-24 < x < 8$ (B) $-8 < x < 24$

(C) $x < -24$ or $x > 8$ (D) $x < 8$

Textbook · Ex. 2 ●○○

2 $|r + 1| \leq 2$

(A) $-1 \leq r \leq 3$ (B) $-3 \leq r \leq 1$

(C) $r \leq -3$ or $r \geq 1$ (D) $r \leq 1$

Textbook · Ex. 3 ●●○

3 $|2c - 1| \leq 7$

(A) $-4 \leq c \leq 3$ (B) $-6 \leq c \leq 8$

(C) $-3 \leq c \leq 4$ (D) $c \leq -3$ or $c \geq 4$

Textbook · Ex. 4 ●●○

4 $|3h - 3| < 12$

(A) $-5 < h < 3$ (B) $-4 < h < 4$

(C) $h < -3$ or $h > 5$ (D) $-3 < h < 5$

Textbook · Ex. 5 ●●○

5 $|m + 4| < -2$

(A) $\varnothing$ (B) $-6 < m < -2$

(C) all reals (D) $m < -6$

Textbook · Ex. 6 ●●○

6 $|w + 5| < -8$

(A) $-13 < w < 3$ (B) $\varnothing$

(C) all reals (D) $w < 3$

Textbook · Ex. 7 ●●○

7 $|r + 2| > 6$

(A) $-8 < r < 4$ (B) $r < -4$ or $r > 8$

(C) $r < -8$ or $r > 4$ (D) $r > 4$

Textbook · Ex. 8 ●●○

8 $|k - 4| > 3$

(A) $1 < k < 7$ (B) $k < -7$ or $k > -1$

(C) $k > 7$ (D) $k < 1$ or $k > 7$

Textbook · Ex. 9 ●●●

9 $|2h - 3| \geq 9$

(A) $h \leq -3$ or $h \geq 6$ (B) $-3 \leq h \leq 6$

(C) $h \leq -6$ or $h \geq 3$ (D) $h \geq 6$

Textbook · Ex. 10 ●●●

10 $|4p + 2| \geq 10$

(A) $-3 \leq p \leq 2$ (B) $p \leq -3$ or $p \geq 2$

(C) $p \leq -2$ or $p \geq 3$ (D) $p \geq 2$

●○○

11 $|x - 3| < 5$

(A) $-8 < x < 2$ (B) $x < -2$ or $x > 8$

(C) $-2 < x < 8$ (D) $x < 8$

●●○

12 $|2x + 4| < 6$

(A) $-5 < x < 1$ (B) $-1 < x < 5$

(C) $-10 < x < 2$ (D) $x < -5$ or $x > 1$

●●●

13 $|x - 5| \geq -1$

(A) $\varnothing$ (B) $4 \leq x \leq 6$

(C) all reals (D) $x \geq 6$

●●●

14 $|x| > 2$

(A) $-2 < x < 2$ (B) $x > 2$

(C) $x > -2$ (D) $x < -2$ or $x > 2$